

ENGINEERING REBOOT 2026

Day 16 Field Manual

Version 1.0 - Week4 Baseline

MISSION BRIEF

Topic: Text File Reading & Writing

Objective: Learn how to persist and retrieve information using text files.

Expected Outcome: Implement a text-file based notes manager that stores and retrieves information from disk.

CONCEPTS

- `open()`
- `read()`
- `write()`
- `append()`
- `with` statement
- File Modes: "r" Read, "w" Write, "a" Append

Example

```
with open('notes.txt', 'w') as file:  
    file.write('Hello World')
```

EXERCISE 01

Filename: `file_write_demo.py`

Objective: Write user-provided content to a text file.

Requirements

- Prompt the user for text input.
- Create a text file.
- Store the entered text in the file.
- Display confirmation that the operation succeeded.

EXERCISE 02

Filename: `file_read_demo.py`

Objective: Read and display content from a text file.

Requirements

- Open an existing text file.
- Read its contents.
- Display the contents.
- Handle an empty file gracefully.

PROJECT / IMPLEMENTATION ASSIGNMENT

Filename: personal_notes_manager.py

Objective: Create a persistent text-file based notes manager.

Functional Requirements

- Add Note
- View Notes
- Exit

Suggested Menu

1. Add Note
2. View Notes
3. Exit

Acceptance Criteria

- Notes are successfully written to disk.
- Notes remain available after program restart.
- Notes can be viewed after being stored.
- Program exits without errors.

VALIDATION MATRIX

- File created successfully
- User input written successfully
- File content read successfully
- Notes persist after application restart
- Notes displayed correctly
- Program exits cleanly

DELIVERABLES

- file_write_demo.py
- file_read_demo.py
- personal_notes_manager.py
- Notes.md
- Journal.md

EVIDENCE CHECKLIST

- Code Evidence
- Output Evidence
- Notes Updated
- Journal Updated

SOURCE CONTROL

- Commit completed

■ Push completed

■ Remote sync verified

PREVIEW OF NEXT TOPIC

Day 17 – File Processing & Data Analysis

Project: `expense_tracker.py`

END OF DOCUMENT